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The Effect of Photobiomodulation in the Management of Leg Ulcers in Sickle Cell Disease: A Systematic Review

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ABSTRACT

Leg ulcer (LU) is a chronic complication of sickle cell disease (SCD), characterised by intense pain, treatment resistance and a high impact on the patient's quality of life (QoL), representing a significant public health challenge. Clinical management involves multifaceted approaches; however, the variability in response to conventional treatments highlights the need for more effective and evidence-based alternatives. Photobiomodulation (PBM) emerges as a promising intervention, with the potential to accelerate tissue healing and reduce pain in patients with SCD-associated LUs. This study conducted a systematic literature review following PRISMA 2020 guidelines, covering relevant databases. Four studies meeting the inclusion criteria were selected, and methodological quality was assessed using the Joanna Briggs Institute tool. Findings indicated that PBM promoted cell growth, accelerated healing and relieved pain, with three studies reporting pain reduction and one study describing a decrease in edema and inflammation. Granulation tissue formation was observed in three of the four studies analysed, and ulcer area reduction was noted in three studies, with one reporting complete healing. Despite the encouraging results, the scarcity of specific studies and the absence of robust randomised clinical trials underscore the need for further research to validate its effectiveness and establish consistent clinical protocols. Expanding these investigations may consolidate PBM as a safe, effective and cost-efficient therapeutic option for patients with SCD-related LUs, contributing to the optimisation of healthcare practices.

1 | Background

Sickle cell disease (SCD) is an inherited genetic disorder characterised by a structural alteration in haemoglobin (Hb), the protein responsible for oxygen transport in the blood. This mutation leads to the formation of haemoglobin S (HbS), causing erythrocytes to acquire a sickle shape, impairing their function and promoting vascular and hemolytic complications [1].

The mutation in the HBB gene replaces a single adenine nucleotide with thymine, leading to the substitution of the amino acid glutamic acid with valine at the sixth position of the β-globin chain. This structural alteration modifies the physicochemical properties of Hb, promoting its polymerisation under low-oxygen conditions. This process results in erythrocyte deformation, causing them to assume a sickle shape and lose their flexibility, making them more adherent to the vascular wall and

Abbreviations: Hb, haemoglobin; HbA, haemoglobin A; HbS, haemoglobin S; Intervention, comparison and outcomes; JBI, Joanna Briggs Institute; LU, leg ulcer; NO, nitric oxide; PBM, photobiomodulation; PICO, patient; PRISMA, preferred reporting items for systematic reviews and meta-analyses; QoL, quality of life; SCD, sickle cell disease; TLR-4, toll-like receptor 4; UHS, Unified Health System (brazilian public health system).

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Key Points

- Methodological rigor through the use of established systematic review guidelines.
- Comprehensive evaluation of clinical outcomes relevant to patients with sickle cell disease.
- Identification of important research gaps and directions for future studies.

susceptible to hemolysis. The sickling process is reversible when the erythrocyte returns to a vascular bed with high oxygen levels. However, due to repetitive cycles of sickling, the erythrocyte membrane becomes irreversibly sickled [1]. SCD manifests clinically when an individual inherits two mutated βS alleles (βS/βS), characterising sickle cell anaemia (SCA).

In heterozygous individuals, HbS can be associated with other mutations, such as haemoglobin C (HbSC) or β-thalassaemia (HbS/β-thal), resulting in clinical conditions of varying severity [1, 2]. Individuals inheriting a single βS allele (HbAS) present with the sickle cell trait, a condition in which HbA and HbS coexist without disease manifestation, although they can transmit the gene to offspring [3].

The clinical complications of SCD are diverse, including vaso-occlusive crises, chronic hemolysis, systemic inflammation and endothelial injury, contributing to ischemic lesions in various organs [4–6]. Among the most debilitating manifestations is the development of leg ulcers (LU), more frequent in SCA individuals, characterised as painful and chronic lesions frequently affecting the medial and lateral malleoli of the ankles, either unilaterally or bilaterally [7, 8].

LUs are skin lesions caused by DF that lead to debilitating clinical conditions with a high risk of recurrence and chronic pain, as well as psychosocial morbidity. Several factors modulate the prevalence and incidence of these ulcers, such as geographic location, socioeconomic conditions, gender, age and genotype. They occur significantly in individuals with FA and less frequently in individuals with SCD [9, 10]. In general, LUs in FD are shallow lesions, without deep tissue involvement, with raised edges and variable size associated with recurrence and chronicity [10, 11]. They have a tissue base consisting of exudate, as a product of vascular and tissue destruction, which may be bloody, purulent or serous. Lesions with purulent exudate are rich in pus, a yellowish fluid of varying density consisting of inflammatory cells, such as neutrophils and macrophages [12].

In addition to physical discomfort, LU is often challenging to heal due to poor local blood circulation and increased susceptibility to infections, significantly impacting patients' quality of life (QoL) [13]. The consequences of this condition extend beyond physical aspects, including mobility restrictions and emotional implications derived from limitations in daily activities, affecting individuals physically, psychologically and socially [8, 12, 13].

Treatment of LUs in SCD encompasses a wide range of clinical management options aimed at improving venous stasis, promoting pain relief and facilitating wound care. Treatment may be topical, including the use of antibiotics, Unna boots, debridement, hydrocolloid dressings or sunflower oil; or systemic (such as improved nutritional support, pentoxifylline, hydroxyurea, blood transfusions and opioids for pain management). Surgical treatment is not commonly used due to a high rate of failure and recurrence [14]. Despite the variety of treatment options, LUs are a debilitating complication of SCD with high recurrence rates and increased morbidity.

Given the limitations of conventional treatments [15], alternative therapeutic approaches have been explored, including photobiomodulation (PBM), also known as low-level laser therapy or low-intensity laser therapy [16, 17]. PBM is a therapeutic modality that utilises light energy to promote biochemical and physiological effects in biological tissues [16–20]. This methodology has been extensively studied for its capacity to modulate cellular processes, playing a crucial role in accelerating wound healing and reducing inflammatory processes [17, 18, 21]. In the context of sickle cell ulcers, PBM emerges as a promising alternative to enhance tissue regeneration and minimise complications associated with impaired wound healing [22, 23] (Figure 1).

This study aims to conduct a systematic review of the scientific literature on the effect of PBM in the management of LUs in individuals with SCD. The research aims to understand the mechanisms of PBM action, evaluate its clinical efficacy and discuss its potential effect on wound healing compared to conventional approaches. Additionally, the study includes an analysis of the methodological quality of selected studies, as well as the editorial quality and impact of the journals in which these studies were published, aiming to ensure the credibility and relevance of the sources used. By exploring the benefits of this therapy, this work intends to contribute to the improvement of therapeutic strategies and provide important insights for future investigations on the use of PBM in the treatment of LU in SCD patients.

2 | Methods

2.1 | Study Design

This study comprises a systematic literature review aimed at evaluating the therapeutic effects of PBM in the treatment of LUs in individuals with SCD. The systematic review aims to collect, evaluate and synthesise the available evidence, providing a comprehensive and unbiased analysis of the existing literature. This approach is particularly relevant in fields with limited data, as it helps identify trends, gaps and the effectiveness of specific interventions.

A systematic review was chosen for its critical assessment of existing case studies, offering a robust understanding of PBM's therapeutic effects. Furthermore, this method standardises the evaluation of methodological quality, reducing potential biases and increasing the reliability of the conclusions.

To ensure consistency and transparency throughout the review, the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines were followed [24, 25]. These recommendations guarantee methodological rigor and completeness, which are crucial for the credibility of the presented results.

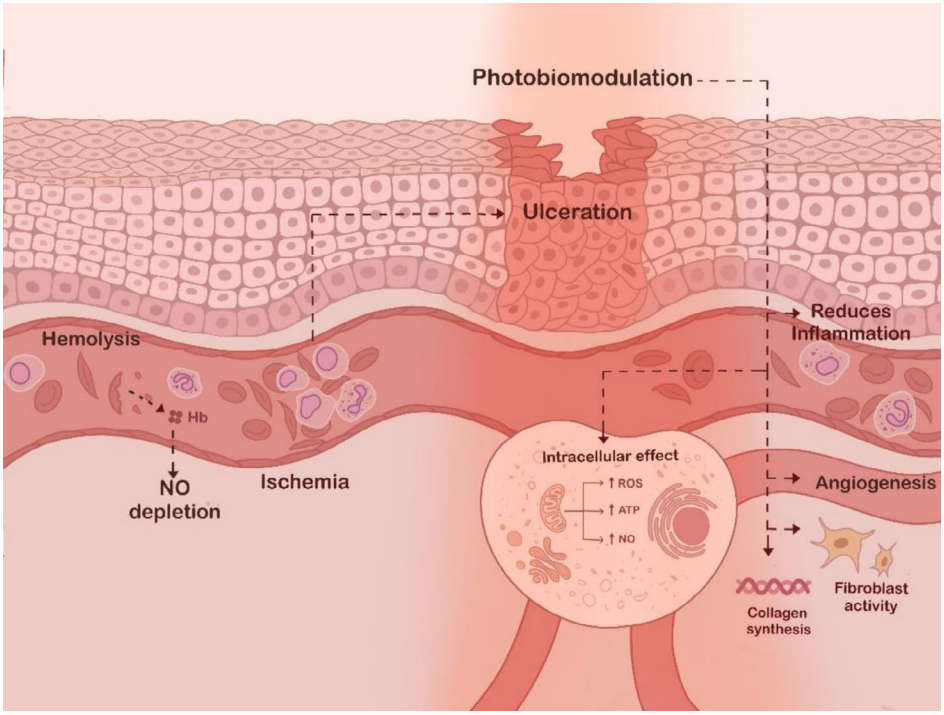


FIGURE 1 | Schematic representation of the biological effects of photobiomodulation on ulcerated tissue. Hemolysis releases free haemoglobin into the tissue microenvironment, which scavenges nitric oxide (NO), leading to NO depletion. Reduced NO bioavailability impairs vasodilation and endothelial function, promoting vascular dysfunction and a pro-inflammatory state. Sustained inflammation further compromises microcirculatory perfusion, resulting in tissue ischemia. Prolonged ischemia disrupts cellular metabolism and tissue integrity, ultimately progressing to epithelial breakdown and ulceration. Photobiomodulation promotes wound healing by modulating cellular and vascular responses. These intracellular effects contribute to reduced inflammation, improved microcirculation and angiogenesis, stimulation of fibroblast activity and collagen synthesis, and overall acceleration of tissue repair. Photobiomodulation also counteracts ischemia-related processes, including hemolysis-associated NO depletion, thereby promoting wound healing and tissue regeneration. Created using Procreate Version 5.4.8.

2.2 | Research Question

The guiding question for this study was structured based on the PICO acronym (Patient, Intervention, Comparison and Outcomes). This model facilitates the clear and targeted formulation of research questions. The target population (P) includes patients with LUs associated with SCD. The intervention (I) consists of PBM application. The comparison (C) was made between PBM and other standard interventions or the absence of treatment. The selected outcomes (O) for evaluation include:

- Ulcer healing
- Pain reduction
- Edema reduction
- Local inflammation control
- Stimulation of granulation tissue formation

2.3 | Search Strategy

An initial scoping search was conducted to assess the feasibility of the systematic review. The official search was performed between October 2023 and March 2024, using the following electronic databases: Pubmed Central, Springer, Wiley and the Virtual Health Library (VHL). To ensure a comprehensive approach, the search was conducted in English, employing descriptors related to 'photobiomodulation', 'laser therapy', 'light therapy', 'wound healing', 'leg ulcer' and 'sickle cell disease'.

Boolean operators, such as 'OR' and 'AND', were utilised to refine and optimise the search. For example, the use of 'AND' facilitated the identification of studies addressing both PBM and LUs simultaneously, while 'OR' allowed for the inclusion of synonyms and related terms, broadening the search scope.

2.4 | Eligibility Criteria

Eligibility criteria were established to ensure the inclusion of relevant and high-quality studies in the systematic review. Given the scarcity of literature on PBM use for LU treatment in patients with SCD, broad criteria were adopted to encompass the largest possible number of pertinent studies.

Inclusion Criteria:

- Studies involving patients diagnosed with SCD.
- Studies investigating PBM as an intervention for LU treatment: Case studies, case reports and prognostic studies that

present data on outcomes such as ulcer healing, pain reduction, edema reduction, granulation tissue presence and local inflammation control.

Exclusion Criteria:

- Studies that did not report PBM results or its effectiveness in LU treatment.
- Articles not available in full text.
- Systematic reviews, meta-analyses and expert opinions, as the focus was directed toward original studies.

2.5 | Study Selection

The selection of studies was conducted in two main phases: initial screening and final selection. In the initial screening phase, the titles and abstracts of the articles identified in the search strategy were evaluated for relevance according to the established inclusion criteria. Studies that did not address PBM in the treatment of LUs in SCD patients were excluded at this stage.

Articles that passed the initial screening were subjected to full-text analysis in the final selection phase. In this stage, the predefined inclusion and exclusion criteria were applied. All studies meeting the eligibility criteria included in the systematic review, enabling a comprehensive analysis of the available evidence. This included evaluating the study methodologies, the population involved, the intervention applied and the reported outcomes.

2.6 | Data Extraction

Data extraction was conducted systematically to ensure consistency and integrity of the collected information. A data extraction form was developed to standardise the process, including all relevant variables such as participant details, intervention characteristics, outcomes and results. The form was pilot-tested on a preliminary set of articles to ensure its effectiveness.

Extracted Variables:

- Study Information: Title, authors, study type, year of publication, journal, country of origin and language.
- Participant Information: Number of participants, age range, gender, race, SCD diagnosis, LU history, duration of LU, location of LU and number of LUs per participant.
- Intervention Details: Type of intervention, equipment used, wavelength, output power, exposure duration per session, irradiated region, session frequency and total number of sessions.
- Outcomes: Ulcer healing, pain reduction, edema reduction, local inflammation control and granulation tissue stimulation.
- Results: Quantitative and qualitative results for the specified outcomes.

2.7 | Data Synthesis

Data synthesis aimed to compile and integrate the information obtained from the selected studies. Initially, a critical reading of each study was performed to identify its main characteristics and quality, as well as the key outcomes related to ulcer healing, pain reduction, edema reduction, local inflammation control and granulation tissue formation. The extracted information was organised into tables for clear visualisation. It was subsequently discussed in an integrative manner, considering the contexts and methodologies of each study.

2.8 | Methodological Quality Assessment

The methodological quality of the included case reports was assessed using the Joanna Briggs Institute (JBI)—Critical Appraisal Checklist for Case Reports. This tool is specifically designed to evaluate the clarity, robustness and transparency of case report presentations, which are essential for interpreting and applying the findings in the context of treatments and interventions for SCD. This evaluation covered:

- Clarity in Case Description
- Methods and Interventions
- Follow-up and Results
- Study Limitations

3 | Results

3.1 | Literature Search Results

The systematic search began with a structured approach, using well-established databases such as Elsevier, LILACS, SciELO and PubMed, Web of Science, SCOPUS, Google Scholar, Chinese Academy of Sciences database. Additionally, other relevant sources were consulted to ensure comprehensive coverage of the topic. During the initial phase, a total of six articles pertinent to the scope of the study were identified. A subsequent screening process identified two duplicate articles, which were excluded from the selected pool of materials. This step was crucial to maintain the integrity and objectivity of the research.

Following this, the remaining articles underwent a full-text review to assess their relevance, adherence to the study's eligibility criteria and data extraction using a specific questionnaire developed by the author. All the analysed articles met the predefined criteria and were considered suitable for inclusion in the study. The selection process and analysis of the materials were documented in a detailed PRISMA flowchart (Figure 2), ensuring transparency and replicability of the study.

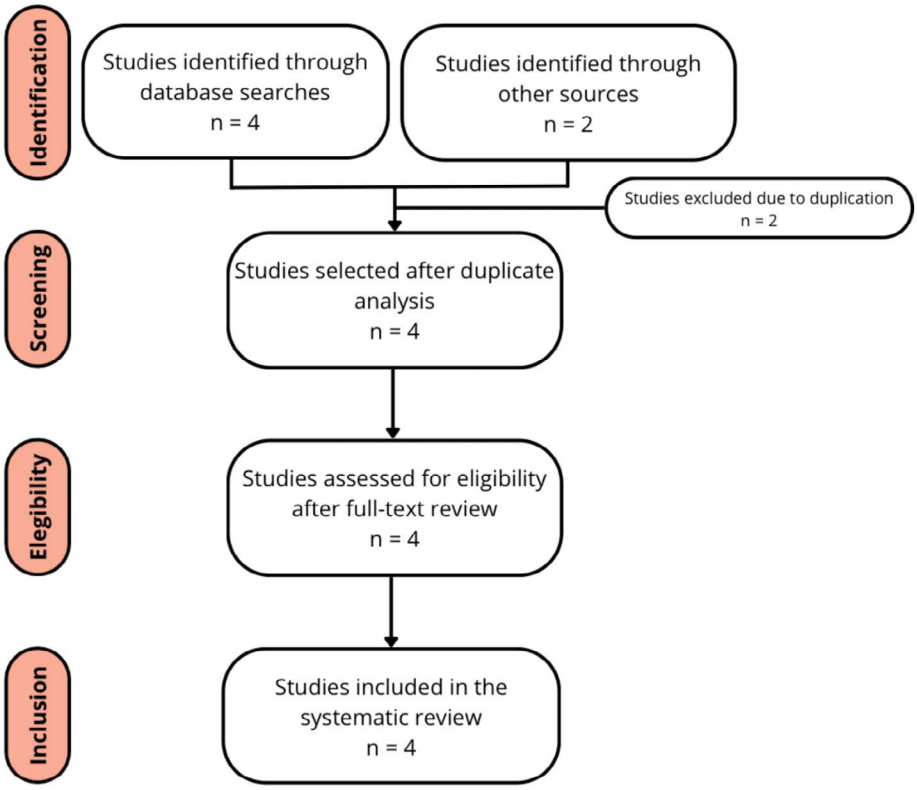


FIGURE 2 | Flowchart of active search and selection of scientific articles. A flowchart describing the process of active search and selection of scientific articles in the systematic review is conducted. The figure outlines the steps of the process, from database searching to the final inclusion of articles, clarifying the selection methodology employed in this study. The flow diagram was elaborated according to the PRISMA 2020 guidelines.

3.2 | Main Findings of the Included Studies

3.2.1 | Study Description

In this systematic review, the studies were organised into tables to facilitate understanding and analysis of the data. The titles of the selected studies are presented in Table 1, providing an overview of the primary investigative works on the use of FBM in treating LU in patients with SCD. Further details, such as study type, year of publication, journal/conference proceedings, country of origin and language of the included articles, are outlined in Table 1.

Among the selected studies, three were case studies and one was a prognostic study, all conducted in Brazil and published in national and international journals and conference proceedings, covering a period from 2007 to 2023.

3.2.2 | Study Characteristics

Details regarding the characteristics of the participants in the case studies included in this systematic review are presented in Table 1. Three participants aged between 28 and 54 years were identified, with the majority being female. Two participants were classified as Black, while the race of one was not specified in the studies. All participants had a diagnosis of HbSS, with one exhibiting the Bantu haplotype. Furthermore, all participants had a history of LUs, with one reporting the ulcer since adolescence. Concerning the ulcer characteristics, two studies did not specify the duration of the ulcerative lesions. The ulcer locations were identified as the right malleolus and above the medial malleolus for most participants, as described in two of the studies, while one study did not detail the exact location of the ulcer for one participant. In all studied cases, only one ulcer per participant was reported.

The detailed characteristics of the participants in the case series were also described in this systematic review (Table 1). A total of six participants were identified, equally distributed between the control group (3) and the experimental group (3). The age range of the participants varied from 20 to 40 years, with an equal distribution between women (3) and men (3). Regarding ethnicity, four participants were classified as mixed race, one as Black and one as White. All participants were diagnosed with HbSS and had a history of LU. The duration of the ulcers ranged from 1 month to 3 years. The predominant ulcer locations were the malleolar, supramaleolar and retromaleolar regions, with the number of ulcers per participant ranging from 1 to 3.

3.2.3 | Description of the Evaluated Outcomes

The outcomes evaluated in the studies included in this systematic review are detailed in Table 2. The studies used different FBM devices, such as Plasmax IV—KLD and PHYSIOLUX DUAL, while the two studies did not specify the devices used. The wavelengths applied were predominantly infrared and red. Regarding output power, only two studies specified the parameters: one used 100 mW, and the other 6 mW. Session durations varied from 10 to 30 min, with two studies not

TABLE 1 | Characteristics of the studies included in this systematic review.

References	Type of study	Year	Journal	Country	Language	Participants	Age	Gender	Race	Diagnosis	History	Duration	Location	No. of LU
Fortuna et al. [23]	Prospective, open-label, uncontrolled	2025	Journal of Tissue Viability	Brazil	English	10	25–56	F = 3 M = 7	Black	HbSS α-thalassaemia	Yes	3–240 months	Medial	2
Manoel et al. [26]	Case report	2023	Hematology, Transfusion and Cell Therapy	Brazil	Portuguese	1	54	F	N/S	HbSS	Yes	N/S	Right malleolus	1
Bonini-Domingos et al. [27]	Case report	2012	Revista Brasileira de Hematologia e Hemoterapia	Brazil	English	1	35	F	Black	HbSS-Bantu	Yes	Since adolescence	Above the medial malleolus	1
Lucena et al. [28]	Prospective, open-label, controlled pilot study	2007	Anais da Faculdade de Medicina da Universidade Federal de Pernambuco	Brazil	Portuguese	EG = 3 CG = 3	20–40	F = 3 M = 3	Black and white	HbSS	Yes	1 month to 3 years	Maleolar, supramaleolar, and retromaleolar regions	1–3

Abbreviations: CG, control group; EG, experimental group; N/S, not specified.

TABLE 2 | Characteristics of the treatment protocol implemented in the studies included in this systematic review.

References	Equipment	Wavelength (nm)	Output power (mW)	Session frequency	Treated area	Session frequency	Total sessions
Fortuna et al. [23]	EMILIGHT	Red (630 nm)	—				60–234
Manoel et al. [26]	—	Infrared (808 nm) Red (660 nm)	100 mW	—	Ulcer bed, edges and saphenous and femoral nerve extension	First and second sessions with a 24-h interval, and subsequent sessions with a 48-h interval	7
Bonini-Domingos et al. [27]	He-Ne Plasmax IV—KLD	Red (655 nm)	—	10–15 min	Ulcer bed, edges and area above the ulcer	Two times per week	5
Lucena et al. [28]	In Gap PHYSIOLUX DUAL	Red (670 nm)	6 mW	30 min.	Surface, edge and surrounding external region of the ulcer	Three times per week (Mondays, Wednesdays and Fridays)	40

specifying the duration. The irradiated regions included the ulcer bed, edges, nerve extension and ulcer margins. Session frequencies ranged from 2 to 3 sessions per week, with the total number of sessions ranging from 5 to 40 across the analysed studies.

3.3 | Methodological Quality Analysis of the Included Studies

The methodological quality of the studies included in this systematic review was assessed using the JBI critical appraisal tools. The analysis aimed to identify the robustness of the studies and the reliability of the findings obtained (Table S1).

3.3.1 | JBI Assessment

Based on the analysis of the included studies, three case studies and one case series were examined using the appropriate JBI critical appraisal tools for case studies. The quality assessment results revealed that the study by Manoel et al. [26] scored 75%, indicating a high level of compliance with the established quality criteria. Conversely, the studies by Bonini-Domingos et al. [27] scored 62.5%, suggesting that although these studies are relevant, their methodologies could be improved to enhance the robustness of the results (Table S1).

Additionally, the case series by Lucena et al. [28] also scored 62.5%, reflecting a need for methodological improvements. These results highlight variability in the quality of the studies, with Manoel et al. [26] and Fortuna et al. [23] studies emerging as the most highly rated, providing a solid foundation for conclusions regarding the effectiveness of FBM in ulcer healing in patients with SCD. The combined analysis of the data suggests that despite the limitations observed in some studies, there is consensus on the benefits of FBM in promoting wound healing and improving outcomes related to pain and edema. These findings underscore the importance of conducting more rigorous research to strengthen the evidence base for this therapeutic approach in treating LUs in SCD patients.

3.4 | Description of the Evaluated Outcomes

The analysis of the outcomes evaluated in the studies included in this systematic review revealed significant findings. A reduction in ulcer area was observed in three out of the four studies analysed, while one study reported complete ulcer healing (Table 3). Additionally, pain reduction was noted in three studies, whereas one did not specify this outcome. In one study, a decrease in edema and inflammation was recorded, indicating a positive response to the treatment. Finally, three studies documented the presence of granulation tissue in the ulcers, suggesting an active healing process, while one study did not specify this finding (Table 3).

Overall, the findings suggest that FBM therapy may promote wound healing, reduce pain and support tissue regeneration in LUs associated with SCD, although further high-quality studies are warranted to solidify these outcomes.

TABLE 3 | Results of the outcomes assessed in the studies included in this systematic review.

References	Primary outcome				Secondary outcomes			
	Ulcer healing							
	Complete epithelialisation		Incomplete epithelialisation		Pain reduction	Edema reduction	Inflammation control	Presence of granulation tissue
	Initial LU area (cm^2)	Final LU area (cm^2)	Initial LU area (cm^2)	Final LU area (cm^2)				
Fortuna et al. [23]	2.0–18.9	0.0	19.8	5.8	Yes	—	Yes	Yes
Manoel et al. [26]	—	—	8.28	6.51	Yes	Yes	Yes	Yes
Bonini-Domingos et al. [27]	—	—	1.5	0.3	—	—	—	—
Lucena et al. [28]	—	—	EG	EG	EG	—	—	EG
			P1 = 14.0	P1 = 9.36	P1 = Yes			P1 = Yes
			P2 = 11.1	P2 = 2.33	P2 = *P/A*			P2 = Yes
			P3 = 5.5	P3 = 4.1	P3 = Yes			P3 = Yes

Abbreviations: EG, experimental group; P1/2/3, patient 1/2/3; P/A, absence of pain before and after treatment.

4 | Discussion

Despite the advances in clinical management, pharmacological treatments and general care related to LUs in individuals with SCD [29], significant challenges remain in treating these complications due to the lack of evidence-based therapeutic guidelines [30]. In this context, PBM Therapy emerges as an innovative and promising therapy that may address the specific demands imposed by the pathophysiology of SCD.

PBM utilises low-intensity light, typically emitted by lasers or LEDs, which is absorbed by cellular chromophores, triggering a cascade of biochemical processes that promote tissue repair, modulate inflammation and enhance microcirculation. These properties are particularly relevant for patients with SCD, whose pathophysiology includes endothelial dysfunction, chronic tissue hypoxia, systemic inflammation and recurrent vaso-occlusion. These factors contribute to the development of hard-to-heal LUs, underscoring the need for therapies that directly target these processes [16, 21].

The studies included in this systematic review reinforce the potential of PBM to address these specific demands. Manoel et al. [26], Bonini-Domingos et al. [27] and Lucena et al. [28] reported significant reductions in ulcer area in patients treated with PBM. Furthermore, the studies by Manoel et al. [26] and Lucena et al. [28] highlighted the formation of granulation tissue, indicating active tissue regeneration. These findings align with previous evidence demonstrating PBM's beneficial effects on wounds of diverse etiologies. Fortuna et al. [23] demonstrated, in a pilot study, that 11 of 17 recurrent ulcers treated with PBM combined with photodynamic therapy (PDT) showed significant improvement within 123 days, including complete healing in nine cases, a reduction in bacterial load, improved tissue quality and decreased pain. These findings support the biological plausibility of PBM and suggest that its combination with PDT may enhance therapeutic effects.

The evolution of an ulcer in an SCD patient, before, during and after 2 months of PBM treatment, is illustrated by Bonini-Domingos et al. [27] (Figure 2). The images depict a substantial reduction in the lesion's extension, accompanied by the formation of granulation tissue and signs of progressive epithelialisation, visually confirming PBM's efficacy in promoting healing. These findings support both qualitative and quantitative reports from the literature, highlighting PBM's potential to significantly improve clinical outcomes for LUs in SCD patients.

A critical characteristic of LUs in SCD patients is tissue hypoxia, exacerbated by hemolytic anaemia and vaso-occlusive crises that impair local blood flow. PBM, by promoting vasodilation and enhancing microcirculation, may increase tissue oxygenation—an essential factor for effective healing. Furthermore, PBM's stimulation of nitric oxide (NO) production is significant in reducing endothelial dysfunction and improving local perfusion, often compromised in SCD patients [16, 17]. Chronic inflammation, a hallmark of SCD, perpetuates tissue damage and delays healing. PBM has demonstrated the ability to modulate inflammatory responses, reducing pro-inflammatory cytokine expression and local edema. These effects, combined with collagen production stimulation and cellular proliferation, may contribute to more efficient tissue repair in SCD patients [18].

Beyond its effects on healing, PBM also provides benefits in managing pain associated with LUs. Studies such as those by Manoel et al. [26], Lucena et al. [28] and Fortuna et al. [23] reported decreased pain intensity following PBM, possibly related to modulation of neural activity and reduction of local inflammation. Despite this, the evidence remains limited, as few studies consistently applied validated pain scales.

However, despite these promising results, this review identified significant limitations in the available literature. Most of the evaluated studies featured small sample sizes and heterogeneous methodological designs, limiting the generalisability of the findings.

Additionally, the lack of randomised controlled clinical trials represents a critical gap in understanding the true effectiveness of PBM in SCD patients. Randomised studies are essential to minimise bias and provide robust evidence, facilitating the formulation of evidence-based therapeutic guidelines. The need for randomised clinical trials has also been described in different populations [31].

The variety of therapy options also reflects the lack of a standardised protocol or guideline for treating the LUs in SCD. Notably, the Brazilian guideline for SCD recommends that the lesions be clean and hydrated, and suggests applying mineral oil and dexamethasone. Additionally, it notes that patients must receive a tetanus vaccine. The estimated cost per year, per event of LUs is 2.368,22 in Brazilian currency for patients seen at the public healthcare system in Brazil [32]. Wound care and patient education are essential for reducing the recurrence and morbidity of LUs in SCD.

The absence of standardisation in PBM application protocols also complicates comparisons across studies. Variations in light parameters, such as wavelength, energy density and treatment duration, may influence outcomes, emphasising the need for greater uniformity in future research. Moreover, multicentre studies with larger sample sizes could help evaluate PBM's impact at different stages of LUs and in more diverse populations.

In summary, PBM appears to be a promising therapeutic intervention for SCD-associated LUs, demonstrating potential in promoting vasodilation, modulating inflammation, stimulating collagen production and accelerating tissue repair. Nonetheless, further robust investigations are required to confirm these findings and explore the specific mechanisms of this therapy in SCD patients.

Although more studies are needed, current evidence suggests that PBM holds promise as a therapeutic strategy capable of addressing the unique pathophysiological challenges of SCD, improving LU healing, reducing pain and potentially enhancing the QoL for these patients.

5 | Conclusions

This systematic review suggests that PBM represents a promising adjuvant therapy for managing LUs in patients with SCD. PBM demonstrated the ability to stimulate cellular growth and accelerate wound healing, contributing not only to clinical improvement but also to enhanced QoL for these patients. However, the limited number of studies specifically addressing PBM application in LUs associated with SCD contrasts with its expanding use in other medical fields. This gap emphasises the urgent need for randomised clinical trials with larger sample sizes to minimise bias and standardise PBM treatment parameters.

In this context, a study involving patients with SCD was conducted to evaluate QoL in individuals with and without LUs, alongside exploring sociodemographic and clinical data. The findings revealed significant associations between factors such as income, occupation and vaso-occlusive episodes with various QoL domains, contributing to a better understanding of the impacts of SCD on patients' lives [33]. These results underscore the importance of considering socioeconomic and clinical conditions in the development of therapeutic strategies for LUs in SCD, including PBM.

To advance the clinical application of PBM, future research should include broader and more diverse populations, integrating different ethnic groups and regions with high SCD prevalence. Detailed characterisation of participants, including ulcer characteristics and treatment history, is crucial for understanding the applicability of results. Additionally, evaluating the combination of PBM with other therapeutic interventions could enhance its beneficial effects.

Fundamental aspects to be addressed in future studies include investigating variables such as optimal dosage parameters, application frequency and therapeutic combinations with other treatments to optimise clinical outcomes. The clinical heterogeneity of SCD patients should also be considered, including the influence of different genotypes and haplotypes on clinical manifestations such as ulcer predisposition, inflammatory response, hemolysis severity and vascular complications. Incorporating genetic analyses may provide deeper insights into individual responses to PBM, enabling the personalisation of therapeutic interventions.

Moreover, there is a pressing need to develop and validate specific tools for bias assessment in case reports, given the current lack of standardised instruments for this type of study. Creating a specific tool would allow for more precise evaluation aligned with the unique characteristics of this research format, contributing to greater reliability and consistency in the conclusions drawn. This tool could include criteria for methodological quality, such as detailed patient description, therapeutic management and analysis of study limitations, ensuring greater transparency in reporting these cases.

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Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Results of the methodological quality assessment of the studies included in this systematic review using the JBI tool.